

Intelligence Features

Demo Roadmap

A demo-focused plan for evolving OPS from a set of operational workflows into an intelligent layer that helps teams capture issues faster, understand context, act, and learn.

TARGET

Demo-ready

TIMELINE

≈ 20 working days

FOCUS

AI-enhanced OPS

Product direction

Capture → Understand → Decide → Execute → Learn

The features below are designed to work as one connected capability set. Voice and evidence capture create better operational data; search and chat make that data accessible; intelligence features connect patterns and history; and the existing OPS Action lifecycle remains the execution backbone.

Demo principles

- Human review stays in the loop. AI suggestions are editable and should not silently change operational records.
- Reuse existing OPS modules, Action lifecycle, permissions and record navigation rather than creating parallel workflows.
- Use realistic curated demo data where production infrastructure is not yet required.
- For demo readiness, favor explainable rules and measurable trends over unsupported predictive claims.

Feature catalogue

Each feature below includes the demo intent, key experience and recommended implementation approach.

FEATURE 01 • CAPTURE • 3–4 DAYS

OPS Voice Capture

A shared voice-first capability across OPS. A user records a short voice note, OPS keeps the audio, transcribes it, converts it into module-specific structured information, and asks the user to review before saving.

- Gemba: voice → structured observation.
- Red Flag: voice → abnormality / incident details.
- Continuous Improvement: voice → problem, idea and benefit.
- Audit: voice → finding / auditor remark.
- Actions: voice → progress or completion update.
- 5S Red Tag: voice → tag reason / issue description.

Demo implementation plan: Start in Gemba, prove the interaction, then reuse the shared capture/transcription layer across modules.

FEATURE 02 • UNDERSTAND • 4–5 DAYS

Global OPS Assistant

A persistent, context-aware chat window available throughout OPS. Users ask operational questions in natural language and receive answers grounded in OPS data, with links back to the relevant records.

- “What needs my attention today?”
- “Show overdue actions in Zone B.”
- “Why was this action created?”
- “Summarize this week’s Gemba issues.”
- Respect module entitlements, user permissions and record-level access.

Demo implementation plan: Begin read-only: find, explain, summarize, compare and navigate. Add write actions only later with explicit confirmation.

FEATURE 03 • UNDERSTAND • 2 DAYS

Universal Search

One cross-module search surface for finding records, people, issues, actions and operational references. It also becomes the retrieval foundation for the OPS Assistant.

- Search Gemba, Red Flag, Audit, CI, Actions, Red Tag and Visual Management.
- Group results by module and record type.

- Direct navigation to the source record.
- Support IDs, titles, equipment/location terms and people.

Demo implementation plan: Build before or alongside the Assistant so chat has a reliable way to retrieve relevant OPS records.

FEATURE 04 • EXECUTE • 2 DAYS

OPS Attention Center

A role-aware queue that answers one question: “What needs my attention now?” It consolidates urgent, overdue and waiting-for-user work across modules.

- Critical and high-severity open items.
- Overdue and due-today Actions.
- Items awaiting review or verification.
- Escalations and unresolved operational records.
- Different views for member, leader, auditor and management roles.

Demo implementation plan: Use existing statuses, due dates and permissions. No AI is required for the first demo version.

FEATURE 05 • DECIDE • 1-2 DAYS

Smart Action Generator

From an observation, finding, Red Flag or improvement record, OPS proposes a corrective action using the source context. The user reviews the proposed title, priority, owner and due date before creating it.

- Suggest action title and description.
- Suggest priority and due date where the source supports it.
- Suggest responsible role/person only when reliable context exists.
- Always preview before committing to the canonical Action store.

Demo implementation plan: Integrate with the existing canonical Action workflow; AI assists creation but does not replace lifecycle rules.

FEATURE 06 • UNDERSTAND • 2 DAYS

Similar / Recurring Issue Intelligence

OPS connects current issues to previous observations, Red Flags, findings and Actions so users can see whether a problem is recurring rather than treating each record in isolation.

- Surface related records around the same asset, location or issue.
- Show recurrence count and recent history.
- Highlight repeated issues after completed corrective actions.

- Link directly to earlier records and Actions.

Demo implementation plan: For demo mode, use deterministic similarity signals and curated related records; production semantic search can come later.

FEATURE 07 • DECIDE • 2 DAYS

Root Cause Assistant

A guided investigation experience for significant or recurring problems. OPS helps the user conduct a structured 5 Why-style analysis and turn the result into corrective and preventive actions.

- Start from a real OPS problem record.
- Guide successive “Why?” questions.
- Summarize the likely root cause for user confirmation.
- Separate immediate correction from preventive action.
- Create Actions only after review.

Demo implementation plan: Launch from recurring/high-impact issues after Similar Issue Intelligence is available.

FEATURE 08 • CAPTURE • 1–2 DAYS

Smart Evidence

Improve the quality of operational evidence, especially Before/After photos. OPS gives the user the original evidence as a reference and helps them capture a useful comparison.

- Show Before photo when capturing After evidence.
- Guidance: same angle, position and relevant area.
- Clear evidence labels and simple mobile flow.
- Later: optional visual comparison assistance.

Demo implementation plan: Demo first with capture guidance and comparison UX. Do not auto-declare an issue resolved from an image.

FEATURE 09 • LEARN • 1–2 DAYS

Daily / Shift Brief

An automatically generated operational briefing that summarizes what happened, what remains open and what deserves attention for a plant, zone or user.

- Yesterday / previous shift summary.
- Critical open items and overdue Actions.
- Items awaiting review.
- Recurring concerns and today’s priorities.

- Role-aware content.

Demo implementation plan: Generate from existing OPS records and Attention Center logic; use AI mainly to turn metrics into a concise narrative.

FEATURE 10 • LEARN • 1-2 DAYS

Connected Record Timeline

A single chronological story of an operational issue, connecting its source, evidence, Action activity, reviews and closure. Related historical issues can be shown alongside it.

- Observation / finding created.
- Evidence captured.
- Action created and assigned.
- Progress / completion evidence.
- Review and verification.
- Related earlier occurrences.

Demo implementation plan: Build on canonical record IDs and existing Action reverse links. Keep the first version read-only and navigational.

FEATURE 11 • LEARN • 2 DAYS

Early Warning Signals

Surface measurable deteriorating patterns before they become larger operational problems. The demo version should use transparent rules and trend calculations rather than claiming predictive AI.

- Rising safety observations in a zone.
- Repeated issues on the same equipment.
- Increasing Action closure time.
- Repeated corrective actions without durable resolution.

Demo implementation plan: Use real calculations over demo data and clearly state the signal that triggered the warning.

FEATURE 12 • UNDERSTAND • 3-4 DAYS

Visual Management Meeting Intelligence

A longer-form voice experience for Tier / Visual Management meetings. OPS records the meeting, produces a summary, identifies decisions and proposes follow-up Actions for review.

- Meeting recording and transcript.
- Summary by topic.
- Detected decisions.

- Proposed Actions, owners and dates where stated.
- Human confirmation before Action creation.

Demo implementation plan: Treat this separately from short Voice Capture because recordings are longer and may contain multiple speakers/topics.

Recommended demo roadmap

Target: a polished end-to-end intelligence demo in approximately 20 working days. The sequence deliberately creates the shared foundations first and then layers intelligence on top.

Days 1–5

Week 1 — Capture & Ask

- OPS Voice Capture in Gemba
- Voice transcription → structured observation
- Universal Search foundation
- Global OPS Assistant shell + read-only Q&A
- Initial Smart Action suggestion path

Demo story: see an issue → speak → OPS structures it → save → ask OPS about it.

Days 6–10

Week 2 — Understand & Act

- Attention Center
- Smart Action Generator refinement
- Similar / Recurring Issue Intelligence
- Connected Record Timeline
- Root Cause Assistant

Demo story: OPS recognizes recurrence → shows history → guides root cause → proposes an Action.

Days 11–15

Week 3 — Management Intelligence

- Daily / Shift Brief
- Smart Evidence
- Early Warning Signals
- Expand Voice Capture to selected modules
- Assistant contextual awareness across major modules

Demo story: management sees what changed, what is deteriorating and what requires attention.

Days 16–20

Week 4 — Meeting Intelligence & Polish

- Visual Management Meeting Intelligence
- Cross-feature UX consistency
- Permission-aware demo paths
- Curated realistic demo scenarios/data
- Regression testing, browser testing and presentation polish

Demo story: meeting → summary → decisions → proposed Actions, connected to the rest of OPS.

Milestones

Milestone	Outcome
Day 5	Voice-first Gemba observation and the first usable Ask OPS experience.
Day 10	Core wow demo: Voice + Assistant + Attention + Similar Issues + Smart Actions + Root Cause.
Day 15	Management layer: Briefs, evidence guidance and early-warning signals.
Day 20	Full integrated demo with meeting intelligence, polished flows and curated end-to-end scenarios.

Demo architecture strategy

For this phase, the objective is a credible partner/client demonstration rather than a production deployment. We can combine the existing browser-local OPS data model with curated realistic records, temporary audio handling and AI API calls. Shared abstractions should still be used where they reduce rework.

- Build Voice Capture once and adapt its structured output by module.
- Build Universal Search as the retrieval layer used by both direct search and the OPS Assistant.
- Use the canonical Action model/store for every proposed or created Action.
- Use existing permission and module-entitlement rules when determining what the Assistant and Attention Center can expose.
- Use deterministic trend/similarity logic where it is more credible than pretending to have a mature predictive model.

- Keep permanent backend migration, enterprise storage, production-scale semantic retrieval and advanced model governance outside this demo scope.

Suggested demo narrative

1	Capture	A supervisor notices a hydraulic leak during a Gemba Walk and records it by voice.
2	Structure	OPS transcribes the voice note and creates a structured observation for review.
3	Connect	OPS identifies similar previous issues on the same equipment and shows the connected history.
4	Investigate	The Root Cause Assistant guides a short investigation.
5	Act	OPS proposes a corrective Action; the user reviews and creates it.
6	Monitor	The issue appears in the appropriate Attention Center until resolved.
7	Ask	A manager asks OPS, “What keeps happening on CNC-04?” and receives a grounded summary with record links.
8	Learn	The Daily Brief / Early Warning view surfaces the recurrence as an operational concern.

Scope note: This roadmap is intentionally demo-ready. Production hardening—persistent audio/file storage, full backend migration, enterprise authentication, scalable retrieval, monitoring, AI evaluation, privacy/governance and reliability work—would be planned separately.